

CSC15S Assignment 6 – PKI and Apache Config

Total: 100pts (due date see Blackboard)

What to turn in: zip and upload all your related files to **Assignment 6** in Blackboard. Your zip should include the following files:

- (1) CSC15Slab6.docx (with all questions completed)
- (2) log_Mission4.txt

Part 1. Dr. Evil's Training Room. Mission 4 – Forrest Gump

[80 pts. Estimated time to complete: 1.5 hrs]

This assignment is named after “*Forrest Gump*”, Dr. Evil’s favorite. Gump and Bubba are launching a new company in Fishing. However, they were tricked by Dr. Evil. The SSL certificate on their website carries the Organization name as **EvilEnterprise**. Your job is to fix the problem with Austin Power, who owns a Certificate Authority center. You need to generate a new Certificate Signing Request (CSR) with the correct registration information (e.g., Organization name as BubbaGumpFish). Then generate (with Austin Power) and deploy the correct certificate to the Apache2 web server. The final product of your assignment is a working **HTTPS** website for BubbaGumpFish.

As usual, you need to start **another terminal** to complete the training. Once you feel ready to submit, please attach **log_Mission4.txt** to Blackboard. The following are some additional hints for each of the challenges.

Challenge	Hint
1. Jenny’s Msg	RSA encryption/decryption using openssl is not covered in lecture. You need to google about it.
2. Bubba-Gump Fish	Q1: use browser to view certificate or use the openssl s_client command; Q2: you need to pass the private key of Gump (even though only the public key generated from it is embedded in the CSR). Why? The CSR itself is signed by the PRIVATE KEY of Gump. That’s why it’s needed.
3. Austin Power’s CA	Use the openssl x509 command introduced in class. Think about why you need both the private key and certificate of Austin Power.
4. Deploy	Q1: you need to use the grep command. Q2: “sudo cp” is the way. Pay attention that now you have multiple private keys/certificates in your working directory. Think about which two you need to copy over to Apache config directory.

Part 2. Background Info of PKI

[20 pts. Estimated time to complete: 20 minutes.

Both questions need no more than 3 sentences of answer. But you’ll need to do some reading.

Q1. What is the **quadratic sieve method** for factoring numbers? [no more than 30 words] (5 pts)

The quadratic sieve is a modern integer factorization algorithm that uses a congruence of squares to find factors, exploiting a specific sequence of numbers that can be factored over a chosen set of small primes.

Q2. What's the **complexity** of quadratic sieve (5 pts)

Sub exponential, $O(e^{\sqrt{\log N \log \log N}})$ where N is number being factored

Q3. Use the complexity formula to explain **why 2048-bit RSA provides only security strength of 112 bits**. You might not end up with the exact number 112, as long as it’s <160 it is fine. (10 pts)

Complexity of factoring a 2048 bit RSA key is roughly equivalent to 2^{112} operations due to the sub-exponential growth of the quadratic sieve, providing an effective security strength of 112 bits